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A knowledge graph-based intelligent planning method for remanufacturing processes of used parts

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ABSTRACT

Intelligent remanufacturing process planning is crucial for the efficient and high-quality remanufacturing of used parts with complex failure characteristics. However, due to the varied failure characteristics of used parts, the diversity of remanufacturing processes, and complex non-linear relationships among remanufacturing process elements, relying solely on mathematical programming or manual empirical is difficult to effectively model and optimise the remanufacturing process planning. To this end, a knowledge graph-based intelligent planning method for remanufacturing processes is proposed to enhance efficiency and quality by combining mathematical programming and knowledge reuse. Firstly, with failure characteristics as decision nodes, a full-element remanufacturing process ontology model is constructed, linking used parts, failure characteristics, and corresponding process plans. The BERT-BiLSTM-CRF model extracts remanufacturing process entities, and a remanufacturing process knowledge graph (RPKG) is constructed. Secondly, an intelligent decision-making model based on graph multi-node path retrieval is proposed. Aim to minimise carbon emissions, time, and cost, combining feature similarity calculations and nearest neighbour search (NNS) to efficiently retrieve the optimal process plan for each failure characteristic. Then, the optimal process plans are merged based on process constraints to create the complete plan. Finally, a concrete case is given to verify the effectiveness and advantages of this method.

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KEYWORDS
Remanufacturing process; full process element; knowledge graph; intelligent planning

1. Introduction

With the increasingly serious problem of global warming, governments and international organisations are actively promoting the development of a low-carbon circular economy and reducing greenhouse gas emissions (Chen et al. 2024; Ehm 2024; Pei et al. 2021). The performance of remanufactured products can reach or even exceed that of prototype new

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products, allowing the value contained in waste resources to be exploited and utilised to the maximum extent (Caterino et al. 2022; Vogt Duberg, Sundin, and Tang 2024). In recent years, the remanufacturing of used parts has become an effective solution for conserving resources and promoting a circular economy (Sahlab et al. 2022).

The remanufacturing process involves operations such as processing methods, sequences, and parameters, which directly affect the carbon emission, processing time, and cost of the remanufacturing process. Consequently, devising efficient and high-quality process plans for remanufacturing is essential to improve the economic and environmental benefits of the remanufacturing industry.

However, remanufacturing is challenging compared to traditional manufacturing (Rizova, Wong, and Ijomah 2020). During the service life of used parts, one or more failure characteristics may occur due to the influence of various factors such as the use environment, load conditions, and material aging (Yankolu and Denizel 2020). There are multiple remanufacturing process methods and operational characteristics for the same failure characteristics. Different remanufacturing process methods and operational characteristics directly influence the carbon emission, time, and cost of the remanufacturing process (Wang et al. 2019). Due to the variability of failure characteristics in used parts and the diversity of remanufacturing methods and operational features, customised remanufacturing process plans are required, significantly increasing the difficulty of remanufacturing process planning. Furthermore, the multitude of remanufacturing process elements and their complex non-linear relationships mean that remanufacturing process planning methods primarily based on human expertise are time-consuming and prone to errors (Jiang, Ding, et al. 2019). It seriously restricts the improvement of economic benefits and the realisation of environmental sustainability of the remanufacturing industry.

In recent years, to obtain the best remanufacturing process scheme under specific targets, many scholars have conducted extensive research on mathematical planning methods. Shakourloo optimises the remanufacturing process by applying stochastic multi-objective objective programming that takes into account profit and cost (Ali Shakourloo 2017). Zheng et al. used the priority-oriented graph to represent the process sequence and developed a mixed integer programming model to derive the best process scheme (Zheng, Liu, and Ahmad 2020). Schepler et al. proposed a heuristic method based on mixed integer programming to solve the process and resource planning problems related to trade-in, disposal and resale of used electronic products (Schepler, Absi, and Jeanjean 2024). Zhang et al. combined the remanufacturing process planning problem and scheduling problem to establish a mixed integer programming model to reduce processing time and carbon emissions (Zhang, Zheng, and Ahmad 2023). Zhang et al. proposed an emergy-based intelligent decision model to determine the optimal remanufacturing process scheme, which could improve economic benefits while reducing environmental impact (Zhang et al. 2021). Lahmar et al. developed a multi-objective mathematical model for hybrid manufacturing and remanufacturing production systems to determine the best plans to minimise total economic costs and carbon emissions (Lahmar et al. 2022). Ke et al. established a mapping model between remanufacturing scheme design (RSD) parameters and carbon emissions and used intelligent algorithms to solve the optimisation model of the optimal RSD parameters (Ke et al. 2023). Stamer and Sauer proposed an algorithm to identify cost-effective remanufacturing process options that can evaluate the best process plans in terms of cost, efficiency, and reliability (Stamer and Sauer 2024). Schepler et al. proposed a refurbishment and remanufacturing planning model for used electronic products (Schepler, Absi, and Jeanjean 2024). To meet the remanufacturing requirements of cost, performance and carbon emission at the same time, Ke et al. proposed an integrated design method of the used product remanufacturing process based on a multi-objective optimisation model (Ke et al. 2024).

The aforementioned studies construct mathematical models and combine optimisation techniques (such as heuristic algorithms) to optimise certain process elements in remanufacturing process schemes, aiming to achieve specific goals such as reducing costs, minimising carbon emissions, or enhancing reliability. These works lay a solid foundation for the development of the remanufacturing field. However, the remanufacturing environment is dynamic, and new failure characteristics, new remanufacturing techniques or equipment may be introduced constantly. When the processing objects or resources change, the models need to be reconstructed and recalculated, which is both time-consuming and labour-intensive (Wu et al. 2021). Additionally, the multitude of elements involved in remanufacturing process schemes and their complex non-linear relationships make it difficult to comprehensively consider the multidimensional process elements affecting time, cost, and carbon emissions in the model, such as remanufacturing methods, equipment, and tools, thus the effectiveness of the models cannot be guaranteed (Wang et al. 2024).

Considering the reusability of process knowledge and its contribution to improving the efficiency of remanufacturing process planning, some scholars conduct research from the perspective of process knowledge reuse. Jiang et al. developed a hybrid method of rough set and CBR for remanufacturing process planning, which improved the efficiency of remanufacturing process planning (Jiang, Jiang, et al. 2019). Huang et al. proposed a parts-level similarity calculation method to realise the reuse of the NC process (Huang et al. 2019). Li et al. proposed an integrated blockchain and CBR-based remanufacturing process planning method to quickly obtain a suitable remanufacturing process plan (Li et al. 2021). He et al. constructed a remanufacturing knowledge model including problem description and solution. Combined with CBR, remanufacturing knowledge can be effectively retrieved and reused (He et al. 2020). Dong et al. constructed a multi-element process knowledge graph and realised the rapid design of the ship parts process through a CBR-based method (Dong et al. 2022). Egbe et al. propose a CBR-based knowledge management framework for fast retrieval and efficient reuse of remanufacturing schemes (Egbe et al. 2023). Wu et al. proposed a new method of fine-grained process retrieval and reuse based on a process component model to improve remanufacturing assembly efficiency (Wu and Liu 2023). Hu et al. extracted and organised remanufacturing disassembly process knowledge from historical data resources, and combined it with rule-based reasoning methods to automatically generate optimal disassembly sequences and schemes (Hu et al. 2024).

The aforementioned methods based on knowledge reuse effectively increase the efficiency of remanufacturing process planning, saving substantial time during the planning phase. However, due to the diversity of used parts and the variability of failure characteristics, unprecedented combinations of failure characteristics often occur, resulting in the inability to find a suitable solution for new problems. Furthermore, previous methods based on knowledge reuse focus on finding historical cases that are most similar to the current problem, but in the case of multi-objective constraints, there is a lack of effective reasoning mechanism to deduce the optimal or satisfactory solution from multiple feasible solutions.

Given the limitations of the above research methods in dealing with remanufacturing process planning, it is necessary to explore a more flexible and intelligent approach to overcome these challenges (Huang et al. 2024). Knowledge graph (KG) is a structured semantic knowledge base consisting of entities, relationships, and attributes. It is an effective tool for knowledge management, which can realise the integration and personalised recommendation of large-scale data with complex structures (Chen, Jia, and Xiang 2020; Nečaský and Stenchlák 2022; Xiao et al. 2023). At present, most scholars apply KG technology to the manufacturing field. Wen et al. proposed a KG-based meta-path question-and-answer method to support process planning (Wen, Ma, and Wang 2023). Zhang et al. combined deep learning with KG to realise macro-process planning (Zhang et al. 2022). Shen et al. proposed a KG-based dynamic knowledge modelling and fusion method for a customised clothing production process (Shen et al. 2023). Guo et al. automatically acquired and integrated multi-source, heterogeneous, multi-layer, and multi-dimensional retired machinery product information and its recessive knowledge through KG to form a gene bank and provide a data basis for remanufacturing process planning (Guo et al. 2024). Aiming at the problems of knowledge accumulation, flexible reuse and fragmented reasoning in machining process design. Hua et al. proposed an intelligent generation method for machining process design based on knowledge graph and deep reinforcement learning (Hua et al. 2024).

Numerous studies demonstrate that KG exhibits robust information integration capabilities, advanced knowledge reasoning abilities, and flexible query-matching mechanisms. In addition, it also has the ability of data visualisation and dynamic update. Given the disparities in the failure characteristics of waste parts and the intricacy of process elements involved in remanufacturing process planning, the KG can not only more effectively capture the process elements associated with remanufacturing process planning and their correlations, but also its highly efficient knowledge reasoning capability and flexible query matching mechanism can support the efficient multi-objective optimisation decision-making process. Therefore, an intelligent remanufacturing process planning method based on KG is proposed. This method integrates the knowledge of remanufacturing processes through KG and combines the advantages of knowledge reuse and mathematical planning methods, which effectively improves the efficiency and quality of remanufacturing process planning. The main contributions of this paper are as follows.

(1) Construction of the full-element RPKG. A full-element RPKG is constructed with failure characteristics serving as decision nodes, integrating initial features of used parts, failure characteristics, and remanufacturing process plans for each failure characteristic (The following are collectively referred to as characteristic process plans). It clearly represents the complex relationships between various remanufacturing process elements in historical cases.
(2) Construction of intelligent decision model based on RPKG. Integrating the advantages of knowledge reuse and mathematical planning methods, an intelligent decision model based on graph multi-node path retrieval is proposed. Aiming to minimise carbon emissions, time, and cost, the model combines feature similarity calculations and NNS to efficiently and intelligently output the optimal characteristic process plans.

The rest of the paper is organised as follows: section 2 presents the overall methodological framework. Section 3 proposes the remanufacturing process domain ontology model and constructs the RPKG. In Section 4, an intelligent planning approach for remanufacturing process based on RPKG is presented. Section 5 presents a case study to validate the proposed approach. Conclusions and outlook are drawn in Section 6.

2. Method overview

To improve the efficiency and quality of customised remanufacturing process planning, an intelligent remanufacturing process planning method based on KG is proposed. The structured integration and intelligent application of historical remanufacturing process cases through KG transforms experience-based manual decision-making into knowledge-based intelligent planning. The overall approach framework, shown in Figure 1, comprises two primary components: Construction of the full-element RPKG and RPKG-based intelligent planning of the remanufacturing process. Detailed steps are as follows.

Step 1: Construction of full-element remanufacturing process ontology model. The remanufacturing process involves multiple factors, including various types of used parts, diverse failure characteristics, complex process flows and equipment requirements, as well as environmental impacts such as carbon emissions. By constructing a comprehensive remanufacturing process ontology model, the relationships between these complex and diverse failure characteristics, process flows, and equipment are structured and standardised, enabling rapid and accurate matching with the most suitable remanufacturing process plan, enhancing decision accuracy and efficiency. Historical remanufacturing process data, sourced from the enterprise database. As shown in Figure 2, primarily consists of three dimensions: characteristic dimension, process information dimension, and key data indicator dimension. Characteristics of used parts include basic attribute characteristics and failure characteristics of used parts. Process information encompasses various process plans and required equipment resources. Key data indicators refer to the time, cost, and carbon emissions required for each process step.

Step 2: Entity extraction and relationship establishment. To accurately extract remanufacturing process entities from historical data, the BIOES labelling method and BERT-BiLSTM-CRF model are used to extract remanufacturing process entities from historical data sets. Then, domain ontologies guide the formation of entity relationship triples, which are stored structurally in Neo4j as RPKG. It integrates dispersed process knowledge, promotes fast information retrieval and intelligent analysis, and provides a powerful and flexible knowledge support framework for coping with diverse failure characteristics.

Step 3: Matching of characteristic process plans based on similarity calculations. According to the basic attribute characteristics and failure characteristics of used parts, the characteristic process plans matching is carried out with the similarity calculation method. During the matching, the edit distance and normalised distance are used to measure string and numeric attribute similarities, ensuring accuracy and adaptability in the matching process, and allowing for rapid identification of highly matched part nodes and failure characteristic nodes for the used part to be processed. Then, starting from the failure characteristic node, all feasible characteristic process plans associated with this failure characteristic are retrieved from the RPKG.

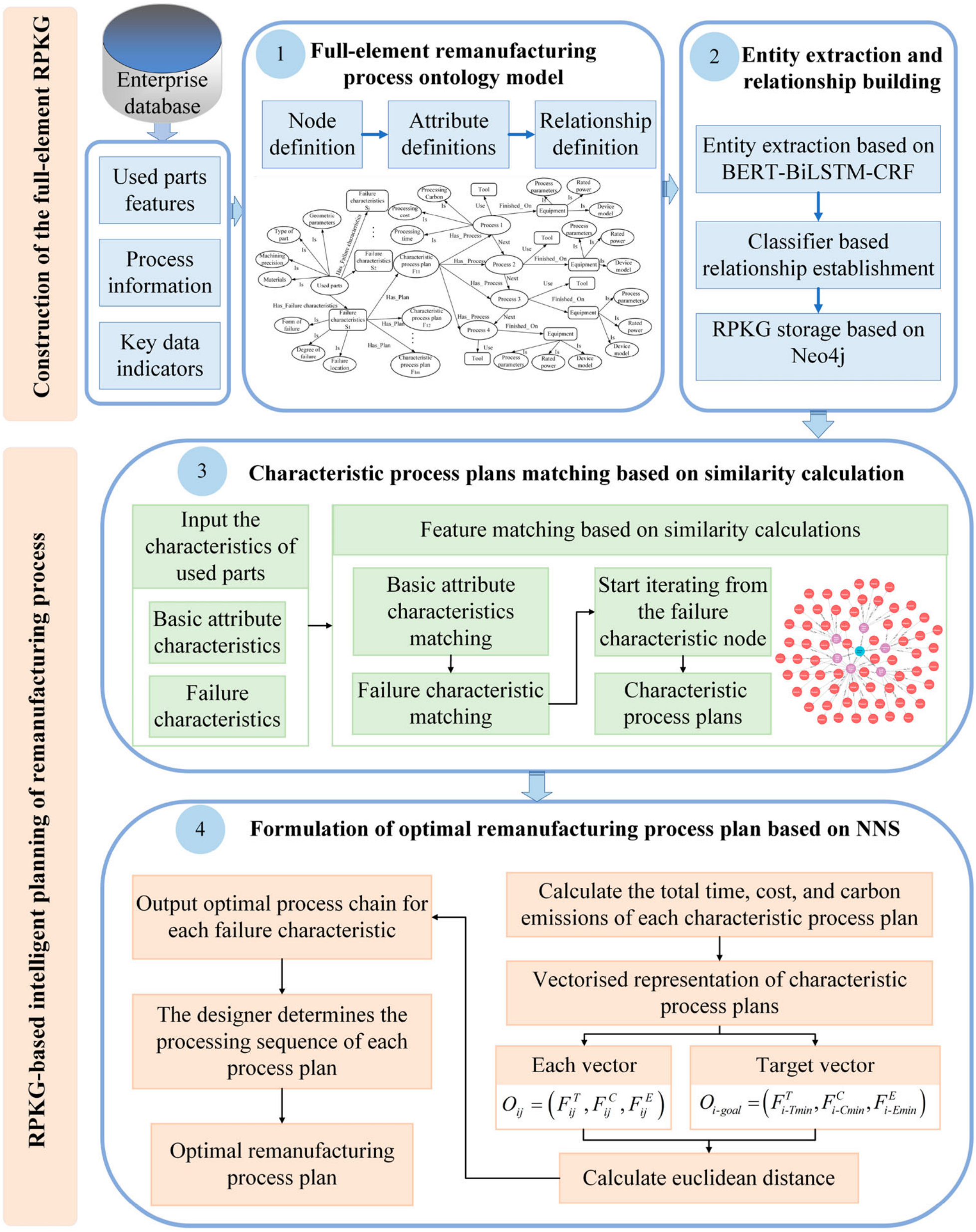

Figure 1. A KG-based intelligent planning framework for the remanufacturing process.

Step 4: Formulation of optimal remanufacturing process plan based on NNS. First, an objective function based on Euclidean distance is constructed, aiming to minimise carbon emissions, time, and cost. Second, the total carbon emissions, time, and cost for each characteristic process plan are calculated and converted into a three-dimensional vector. The target vector is determined based on the minimum processing time, lowest processing cost, and least carbon emissions across all characteristic process plans. Then, the Euclidean distance between each vector and the target vector is computed, and the characteristic

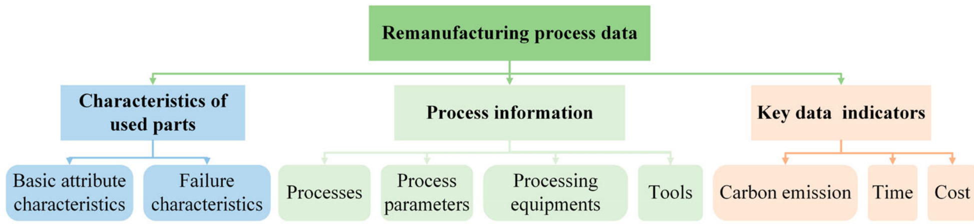

Figure 2. Three dimensions of data.

process plan with the smallest distance to the target vector is outputted. For used parts with multiple failure characteristics, designers combine the retrieved optimal characteristic process plans according to inter-process constraints to form a complete remanufacturing process plan.

3. Construction of the full-element RPKG

3.1. Ontology model of full-element remanufacturing process

At the engineering semantic level, ontology models are often used to formalise knowledge and its domain associations (Nuñez and Borsato 2018; Zangeneh and McCabe 2020). The ontology construction process is essentially a formal representation of a knowledge structure that allows for the explicit definition of entity nodes, relationships between entities, and properties of the entities themselves. Nodes, as the basic constituent units in an ontology, represent specific entities or abstract concepts. Interactions between entities are described through relationships. Attributes are used to describe the essential characteristics or specific properties of entities. By defining nodes, relationships, and attributes, the aim is to construct a unified semantic framework for in-depth understanding and standardised interpretation of remanufacturing processes.

(1) Definition of nodes, attributes, and relationships. Within the context of remanufacturing process workflows, knowledge nodes are categorised into four types: Feature nodes (F_Entity), Process nodes (P_Entity), Machine nodes (M_Entity), and Tool nodes (T_Entity). Table 1 describes the definitions. Feature nodes aim to distinguish and document the initial features and failure characteristics of used parts, enabling precise localisation and description of information during the remanufacturing process. Process nodes concentrate on sequences of process operations and their detailed steps, laying the groundwork for the logical construction of process flows. Machine nodes pertain to various mechanical equipment employed in the processing, ensuring comprehensive coverage of the physical execution aspect. Tool nodes encompass tools and auxiliary devices used in manufacturing.

Among them, used part nodes, part failure characteristic nodes, process nodes, and equipment nodes each have some attributes describing the characteristics, properties, or conditions of their corresponding entities. Detailed definitions are provided in Table 2.

In the context of remanufacturing processes, interactions amongst nodes and between node attributes are classified into three principal categories: Feature Relations (Rf), Demand Relations (Rd), and Sequence Relations (Rs). Feature Relations (Rf) primarily delineate the attributes of used parts, furnishing crucial insights into the fundamental state of the remanufacturing object. Demand Relations (Rd) chiefly articulate the connections between failure

Table 1. Definitions of different types of knowledge nodes.

Node type	Node definition	Specific examples
F_Entity	Type and name of used parts	Worm gears, machine tool spindles, etc
	Failure characteristics	Abrasion, cracks, corrosion, etc
P_Entity	Characteristic process plans	A_1, A_2, A_3, etc
	Process	Electroplating, laser cladding, etc
M_Entity	The equipment that completes this process	Electroplating brush machine, laser cladding equipment, etc
T_Entity	The tools required to complete this process	Turning tools, milling cutters, diamond grinding wheels, etc

Table 2. Node attributes.

Node	Node properties
Used parts node	part type, geometric parameters, machining accuracy, material
Failure characteristics node	form of failure, degree of failure, location of failure
Process node	time, cost, carbon emissions
Equipment node	equipment type, rated power, process parameters

Table 3. Node relationship definitions and their descriptions.

Relationship type	Relationship Definition	Name of relations	Description of relations
Feature Relations(R_f)	The failure characteristics of the part	Has_Failure characteristics	$A \xrightarrow{\text{Has\_Failure characteristics}} B$
	The property value of the node	Is	$A \xrightarrow{\text{Is}} B$
Demand Relations(R_d)	The characteristic process plan for a failure characteristic	Has_Plan	$A \xrightarrow{\text{Has\_ Plan}} B$
	Processes in a characteristic process plan	Has_Process	$A \xrightarrow{\text{Has\_ Process}} B$
	The equipment used to complete this process	Finished_On	$A \xrightarrow{\text{Finished\_ On}} B$
	The tool is used to complete this process	Use	$A \xrightarrow{\text{Use}} B$
Sequence Relations(R_s)	The sequence between operations	Next	$A \xrightarrow{\text{Next}} B$

characteristics and characteristic process plans, characteristic process plans and processes, processes and machinery, as well as tooling, thereby elucidating the inherent logic of technological requirements and resource allocation in the remanufacturing process. Sequence Relations (Rs) strictly dictate the temporal sequence logic of process execution in remanufacturing, mandating the completion of one step before proceeding to the next. Stemming from a meticulous analysis of actual remanufacturing workflows, seven types of relationships between knowledge nodes are identified, with explicit definitions detailed in Table 3.

(2) Construction of knowledge ontology model of remanufacturing process. Considering the peculiarities of the remanufacturing process for used parts and in light of the definitions for nodes, relations, and attributes outlined above, a remanufacturing process ontology model is constructed, as depicted in Figure 3. The full-element remanufacturing process ontology, with used parts as the top-level concept, integrates nodes, attributes,

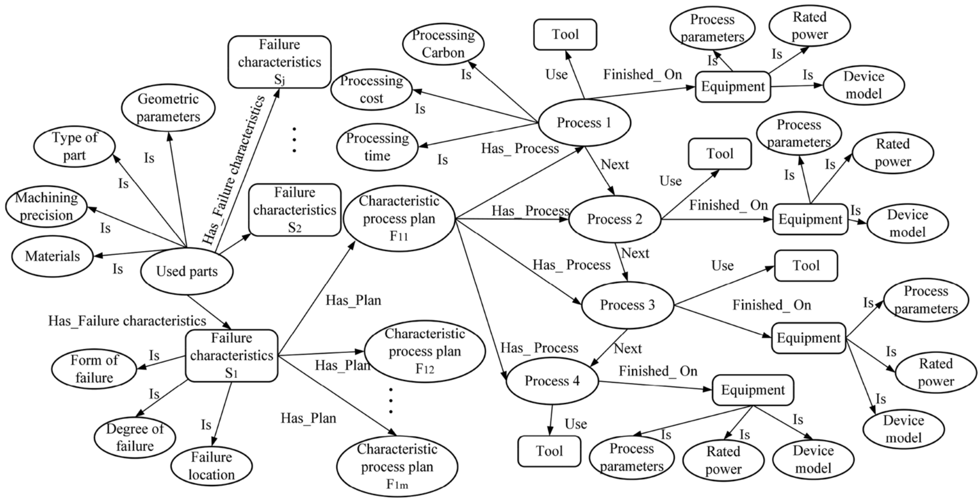

Figure 3. Full-element remanufacturing process ontology.

and relationships pertinent to characteristics of used parts, remanufacturing process information, and critical data indicators. This linkage of knowledge nodes related to the entire remanufacturing process forms a semantically rich network structure.

3.2. Entity extraction and relationship establishment

Entity identification and data preprocessing are performed using Named Entity Recognition (NER) technology. Then, seven relationships between entities are established through the classifier. Finally, the Neo4j database is used for efficient knowledge storage, query, and visualisation, to comprehensively manage and utilise complex information in the field of remanufacturing.

NER is the most commonly used method to automatically identify named entities from a raw corpus (Li et al. 2020). Given the diversity of document formats in the remanufacturing domain, which may lead to complex and differently formatted data structures, data preprocessing is required to improve data quality. First, data in other formats are converted to readable content, and special symbols or unrecognisable characters are removed. Then, periods are used as separators to divide the document into multiple sentences. Finally, entities in the text are annotated to clarify entity boundaries and distinguish entity types to better perform entity extraction tasks. Since the remanufacturing process domain contains several single-character entities, such as 'Axis' and 'Turning', the remanufacturing process entities are labelled using the BIOES labelling method. The BIOES markup method uses B (Beginning), I (Inside), E (End), and S (Single) labels to clearly define where entities in text start and end, and O (Outside) labels to label words that do not belong to any entity. At the same time, to effectively extract and characterise remanufacturing process domain knowledge, it is necessary to define the category of entities. BIOES annotation rules and entity class definitions are shown in Table 4. These annotated data are used as the basis for training the NER model.[open-strick]

Table 4. Entity labelling rules.

Entity type	Typical example	BIOES labelling
Part entity(Parts)	Axis	Axis /S-par
Invalid feature entity(Failure)	Tooth flank wear	Tooth /B-fai flank / I-fai wear /E-fai
Process method entity(Method)	Cold welding	Cold /B-met welding /E- met
Equipment entities(Equipment)	Plating brush machine	Plating /B-equ brush /I- equ machine /E- equ
…		
Nonprocess entities	Include	Include /O

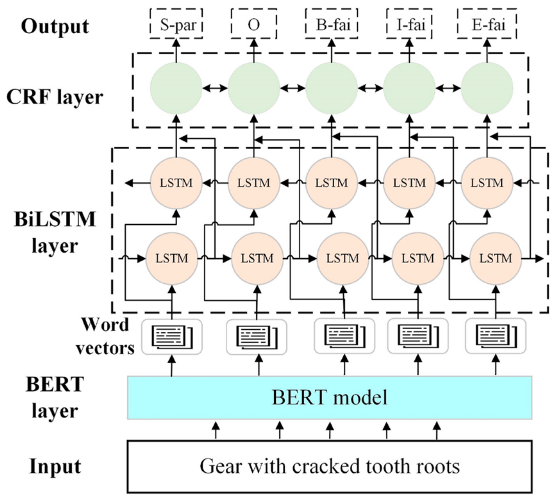

Figure 4. Structure of the BERT-BiLSTM-CRF model.

[close-strick]In the realm of remanufacturing, an open corpus is currently lacking. To address the challenges posed by limited training datasets and polysemy, this study employs a deep learning approach based on the BERT-BiLSTM-CRF model for NER, with the workflow illustrated in Figure 4. First, a pre-trained BERT model serves as an input feature extractor to capture entity information in the remanufacturing process knowledge text. Built upon multi-layer Transformer encoders, BERT boasts robust semantic modelling capabilities, generating context-aware vector representations for each word or token. These representations are then channelled into a BiLSTM layer for further processing, thereby integrating entity information from remanufacturing procedure descriptions into sequential tagging tasks. The extracted features are processed by the BiLSTM layer, comprised of twin LSTM layers processing sequences in forward and reverse orders, respectively. Lastly, a CRF layer is introduced to execute sequence labelling, leveraging transition and emission matrices. By considering global constraints among labels, the CRF layer, trained on annotated data, optimises entity recognition accuracy and consistency.

When establishing relationships between entities, reuse the seven relationships defined when building the pattern layer. By constructing classifiers to identify and establish relationships among data layer entities, relationship recognition is transformed into a supervised classification problem. The original text and labels of each entity are input into the classifier, which outputs the predicted probabilities of the seven relationship types and selects the relationship category corresponding to the greatest probability.

The storage and visualisation of KG is an important part of managing and utilising KG. Neo4j provides a graph-based data model that can efficiently store and manage complex entities, relationships, and attributes. Designed with the performance requirements of large data sets in mind, Neo4j can efficiently process complex knowledge graphs containing millions or even billions of nodes and edges. Its built-in indexing mechanism and optimised query engine ensure fast response times on massive amounts of data. In addition, Neo4j provides a rich set of graph algorithms and graph analysis tools to help users effectively query and analyze complex data in KG, further improving its management and analysis capabilities under large-scale data sets (Cheng et al. 2024).

4. Intelligent decision-making model based on graph multi-node path retrieval

Due to the difference in the types and failure characteristics of used parts, the previous overall process reuse strategy can't achieve good results. Furthermore, there has been a lack of reasoning about historical process knowledge, which makes it difficult to effectively balance these objectives under multiple constraints and provide satisfactory solutions. Therefore, based on the RPKG, an intelligent decision-making model based on multi-node path retrieval in graphs is proposed. This model aims to minimise carbon emissions, time, and cost by utilising graph path retrieval technology, combined with feature similarity calculation and NNS, to achieve intelligent retrieval of the optimal remanufacturing process plan for used parts. This enhances the adaptability and accuracy of the process schemes while also considering environmental and economic benefits. As shown in Figure 5. The model primarily consists of two parts: characteristic node matching based on similarity calculation and optimal process plan retrieval based on NNS.

4.1. Characteristic node matching based on similarity calculation

To improve the accuracy of characteristic process plan matching, the basic property characteristics and failure characteristics of used parts to be processed are considered at the same time. The detailed process is shown in Figure 6. Firstly, according to the basic attributes characteristics of used parts to be processed, parts nodes that meet the similarity conditions are retrieved from RPKG through similarity calculation. The similarity threshold for the basic attribute features is set to ϵ_1. Subsequently, based on the retrieved parts nodes, failure characteristic nodes that satisfy similar conditions are further searched according to the failure characteristics of the used parts to be machined. The similarity threshold for the failure feature attributes is set to ϵ_2. When the failure characteristic node similar to the used parts to be processed is found, the RPKG is traversed by using the retrieved failure characteristic node as the starting node through the match query statement, to obtain all characteristic process plans corresponding to the used parts to be processed.

Similarity is divided into local and global similarity, local similarity is between two attributes and global similarity is between two entities. When calculating local similarity, there are two different types of features: features of numeric type and features of string type. To improve the matching accuracy, different similarity calculation methods are used to calculate the similarity of the two types of attribute features. For numerical type features,

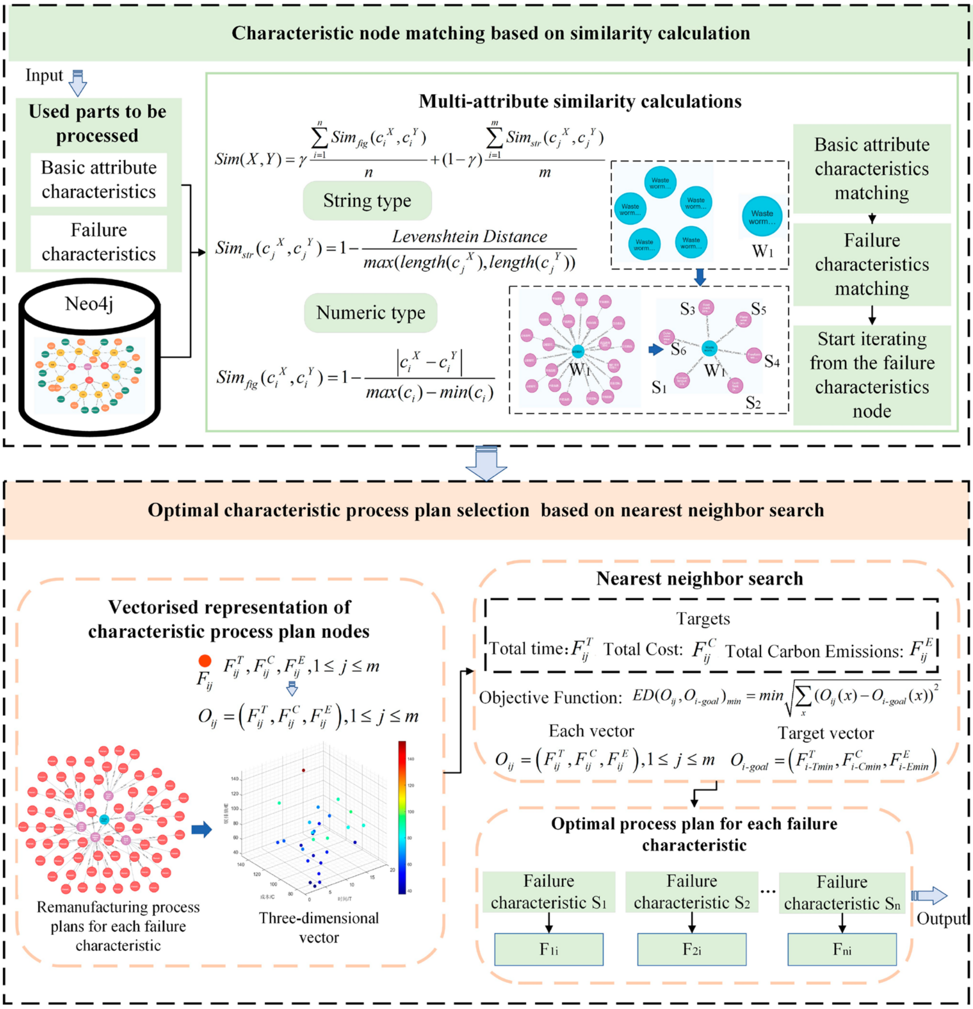

Figure 5. Intelligent decision-making model based on graph multi-node path retrieval.

the similarity calculation formula is as follows.

$$Sim_{fig}(c_i{}^X, c_i{}^Y) = 1 - \frac{|c_i{}^X - c_i{}^Y|}{max(c_i) - min(c_i)} \tag{1}$$

Where X is the basic attribute characteristic entity or failure characteristic entity of the used part to be processed, Y represents the existing part node or failure feature node in the RPKG, $c_i{}^X$ and $c_i{}^Y$ represent the numerical value type of the features of X and Y, $max(c_i)$ and $min(c_i)$ are respectively the maximum and minimum values of attribute c_i at all part nodes.

For string-type features, the edit distance similarity is employed as the computation method. Edit distance is the minimum number of edit operations required to convert one string to another. Editing operations include inserting, deleting, and replacing characters. The smaller the edit distance, the greater the similarity between the two strings. The edit

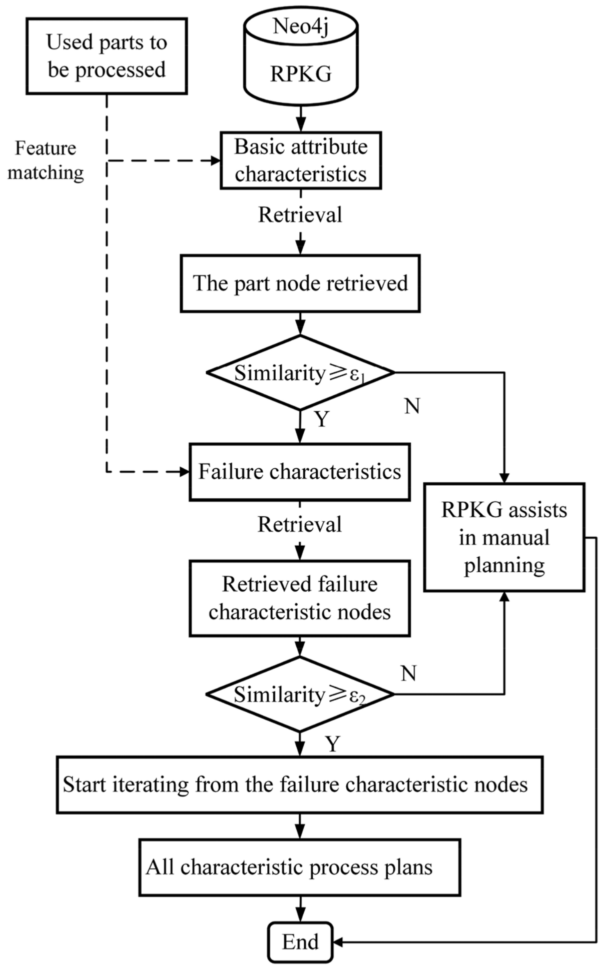

Figure 6. Characteristic matching.

distance similarity formula is as follows.

$$Sim_{str}(c_j{}^X, c_j{}^Y) = 1 - \frac{Levenshtein\ Distance}{max(length(c_j{}^X), length(c_j{}^Y))} \tag{2}$$

Where $c_j{}^X$ and $c_j{}^Y$ denote the jth string type feature of X and Y respectively, Levenshtein Distance refers to the edit distance between the two strings, $length(c_j{}^X)$ and $length(c_j{}^Y)$ are the lengths of the two strings, $max(length(c_j{}^X), length(c_j{}^Y))$ is the maximum value of the two strings lengths.

Therefore, the global similarity between two entities is shown in equation (3).

$$Sim(X, Y) = \gamma \frac{\sum_{i=1}^{n} Sim_{fig}(c_i{}^X, c_i{}^Y)}{n} + (1-\gamma)\frac{\sum_{i=1}^{m} Sim_{str}(c_j{}^X, c_j{}^Y)}{m} \tag{3}$$

Where $\frac{\sum_{i=1}^{n} Sim_{fig}(c_i{}^X, c_i{}^Y)}{n}$ and $\frac{\sum_{i=1}^{m} Sim_{str}(c_j{}^X, c_j{}^Y)}{m}$ are the average similarity between X and Y for numeric type features and string type features respectively, n is the number of numeric type

Table 5. The calculation process of used parts similarity.

Algorithm 1: Similarity calculation of used parts

Input: Used part nodes in the graph database (w: used_parts); Target numeric attribute value (c_i_target); Target string attribute value (c_j_target);
Output: The most similar used part node; Total_similarity; Numeric attribute similarity; String attribute similarity
1: Initialise variables
allPlans: Empty list; min c_i^y: Infinity; max c_i^y:c_i^y Negative infinity; c_i_target: Target numeric attribute value; c_j_target: Target string attribute value; n: Number of numeric attributes; m: Number of string attributes
2: Iterate over all used part nodes w_i
if w_i.name contains 'The name of the used part to be processed'
Calculate the numeric attribute c_i^y
Add w_i, c_i^y,c_j^y to allPlans list
Update min c_i^y and max c_i^y
3: For each plan in allPlans, extract w_i, c_i^y, and, c_j^y
4: Calculate similarity
Numeric attribute similarity c_i_similarity $= \sum_{i=1}^{n} 1.0 - \mathrm{ABS}(c_i^y - c_i\_\mathrm{target}) / (\max c_i^y - \min c_i^y)$
String attribute similarity c_j_similarity $= \sum_{j=1}^{m} \mathrm{LevenshteinSimilarity}(c_j^y, c_j\_\mathrm{target})$
Total similarity $= (c_i\_\mathrm{similarity} + c_j\_\mathrm{similarity}) / (n + m)$
Add w_i, Total_similarity, c_i_similarity, c_j_similarity to results list
5: Sort the results list by Total similarity in descending order
6: If there is a Total_similarity $\geq \epsilon 1$ of a scheme
Return the most similar used part node and its related similarities

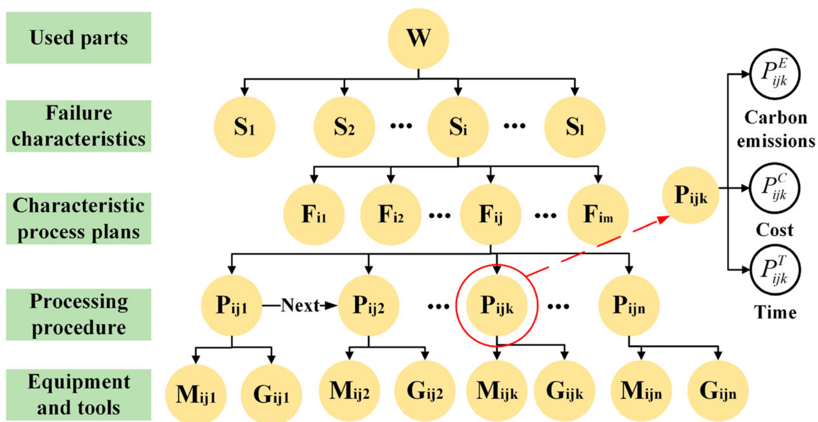

Figure 7. Schematic diagram of RPKG.

features, m is the number of string type features and γ is the weight coefficient of numeric type feature similarity in total similarity, and in this paper, γ is set to 0.5. The pseudocode for searching the most similar used parts nodes for the used parts to be processed is shown in Table 5. The retrieval process of the failure characteristic nodes is consistent with that of the used parts nodes.

4.2. Optimal characteristic process plan selection based on NNS

As shown in Figure 7, the failure characteristics of the used part W are $S = \{S_1, S_2, \ldots, S_i, \ldots, S_l\}$. The set of characteristic process plans against the failure characteristic S_i are $F_i = \{F_{i1}, F_{i2}, \ldots, F_{ij}, \ldots, F_{im}\}$, m is the total number of characteristic process plans. The processes in F_{ij} are $P_{ij} = \{P_{ij1}, P_{ij2}, \ldots, P_{ijk}, \ldots, P_{ijn}\}$, the time, cost, and carbon emissions to complete this process are stored at each process node. The equipment to complete each process is $M_{ij} = \{M_{ij1}, M_{ij2}, \ldots, M_{ijk}, \ldots, M_{ijn}\}$, and the tool used is $G_{ij} = \{G_{ij1}, G_{ij2}, \ldots, G_{ijk}, \ldots, G_{ijn}\}$, where P_{ijk} is the kth process in F_{ij}, M_{ijk} is the equipment used to complete process P_{ijk}, and G_{ijk} is the tool used to complete process P_{ijk}. In plan F_{ij}, the completion time of each process is $P^T_{ij} = \{P^T_{ij1}, P^T_{ij2}, \ldots, P^T_{ijk}, \ldots, P^T_{ijn}\}$, the cost is $P^C_{ij} = \{P^C_{ij1}, P^C_{ij2}, \ldots, P^C_{ijk}, \ldots, P^C_{ijn}\}$, and the carbon emission generated is $P^E_{ij} = \{P^E_{ij1}, P^E_{ij2}, \ldots, P^E_{ijk}, \ldots, P^E_{ijn}\}$, where P^T_{ijk},P^C_{ijk}, and P^E_{ijk} are the time, cost, and carbon emission of using tool G_{ijk} to complete process P_{ijk} on equipment M_{ijk} respectively.

The total time, total cost, and total carbon emissions required to complete the characteristic process plan F_{ij} are shown in equation (4).

$$\begin{cases} F^T_{ij} = \sum_{k=1}^{n} P^T_{ijk} \\ F^C_{ij} = \sum_{k=1}^{n} P^C_{ijk} \\ F^E_{ij} = \sum_{k=1}^{n} P^E_{ijk} \end{cases} \tag{4}$$

Each characteristic process plan is converted into a three-dimensional space vector according to its processing time, cost, and carbon emission. If the processing time, cost, and carbon emission of characteristic process plan are $(F^T_{ij}, F^C_{ij}, F^E_{ij})$, respectively, its corresponding vector is $O_{ij} = (F^T_{ij}, F^C_{ij}, F^E_{ij}), 1 \leq j \leq m$. As shown in Figure 8, the vector $O_{i-Tmin} = (F^T_{i-Tmin}, F^C_{i-Tmin}, F^E_{i-Tmin})$ corresponds to the characteristic process plan F_{i-Tmin} with the shortest processing time, the vector $O_{i-Cmin} = (F^T_{i-Cmin}, F^C_{i-Cmin}, F^E_{i-Cmin})$ corresponds to the characteristic process plan F_{i-Cmin} with the lowest processing cost, and the vector $O_{i-Emin} = (F^T_{i-Emin}, F^C_{i-Emin}, F^E_{i-Emin})$ corresponds to the characteristic process plan F_{i-Emin} with the lowest processing carbon emission. For failure characteristic S_i, the shortest processing time of all characteristic process plans is F^T_{i-Tmin}, the lowest cost is F^C_{i-Cmin}, and the lowest carbon emission is F^E_{i-Emin}. Regard F^T_{i-Tmin},F^C_{i-Cmin},and F^E_{i-Emin} as the target vector O_{i-goal}, then $O_{i-goal} = (F^T_{i-Tmin}, F^C_{i-Cmin}, F^E_{i-Emin})$.

Then, the euclidean distance between vector O_{ij} and the target vector O_{i-goal} is calculated by NNS. The euclidean distance is calculated by the following formula.

$$ED(O_{ij}, O_{i-goal}) = \sqrt{\sum_x (O_{ij}(x) - O_{i-goal}(x))^2} \tag{5}$$

Where, x is the time, cost, and carbon emission. If $ED(O_{ij}, O_{i-goal})$ is the smallest, then F_{ij} corresponding to O_{ij} is the optimal characteristic process plan considering time, cost, and carbon emission.

The objective function is defined as follows.

$$ED(O_{ij}, O_{i-goal})_{min} = min\sqrt{\sum_x (O_{ij}(x) - O_{i-goal}(x))^2} \tag{6}$$

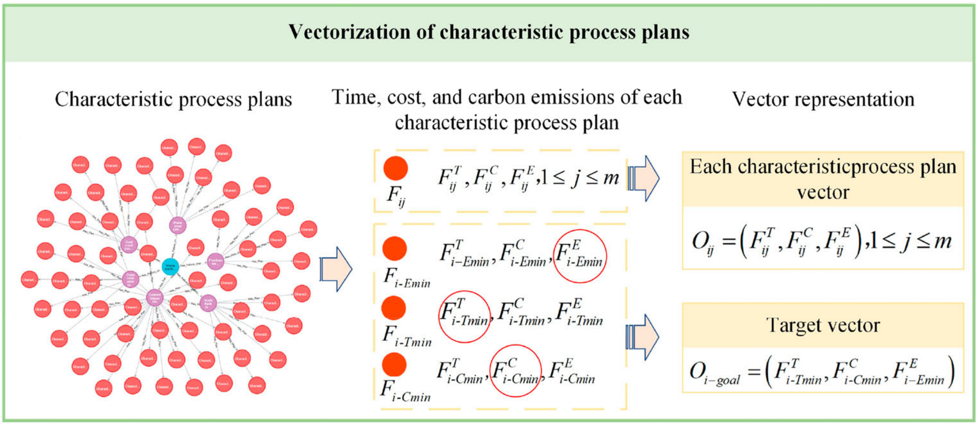

Figure 8. Vectorisation of characteristic process plans.

To find the best characteristic process plan, NNS is implemented by cypher language in Neo4j. To ensure efficient retrieval, after matching the basic attribute characteristics and failure characteristics of the to-be-processed used part to the corresponding used part and failure characteristic nodes in the knowledge graph, input is the most matched used part and failure characteristic nodes. Subsequently, the feasible remanufacturing process plans, equipment, tools, and other related nodes corresponding to this failure characteristic node are traversed, rather than traversing the entire graph, which significantly reduces the search space. Finally, under multiple constraints, the optimal solution is output by measuring the distance between the time, cost, and carbon emission of each plan and their respective minimum values using the Euclidean distance. The search process is shown in Figure 9, and the detailed steps are as follows.

Step 1: The process nodes in each characteristic process plan are traversed and the total time (T), total cost (C), and total carbon emission (E) of each characteristic process plan are obtained respectively. Assuming that each remanufacturing process scheme has m process nodes and a total of n remanufacturing process plans, the total time complexity is O(n × m).

Step 2: The calculated total time, total cost, and total carbon emissions are updated to the characteristic process plan node. The time complexity of this Step is included in Step 1.

Step 3: The total time, total cost, and total carbon emissions at each characteristic process plan node are collected into a list. Going through n remanufacturing process plans and storing them in a list has a time complexity of O(n).

Step 4: Find the minimum total time F^T_{i-Tmin}, minimum total cost F^C_{i-Cmin}, and minimum total carbon emissions F^E_{i-Emin} in the list to get the target vector O_{i-goal}. Time complexity is O(n).

Step 5: Calculate the euclidean distance $ED(O_{ij}, O_{i-goal})$ between O_{ij} and O_{i-goal}. Traversing all n characteristic process planning nodes and calculating distances, the time complexity is O(n).

Step 6: Sort the calculated results in Step 5 from small to large. Returns the characteristic process plan closest to O_{i-goal}, as well as its process, equipment, and tool nodes. The time complexity of the sorting operation is O(nlogn).

In summary, if the number of process steps for each remanufacturing process plan is m, the total time complexity is O(n × m + 3n + nlogn).

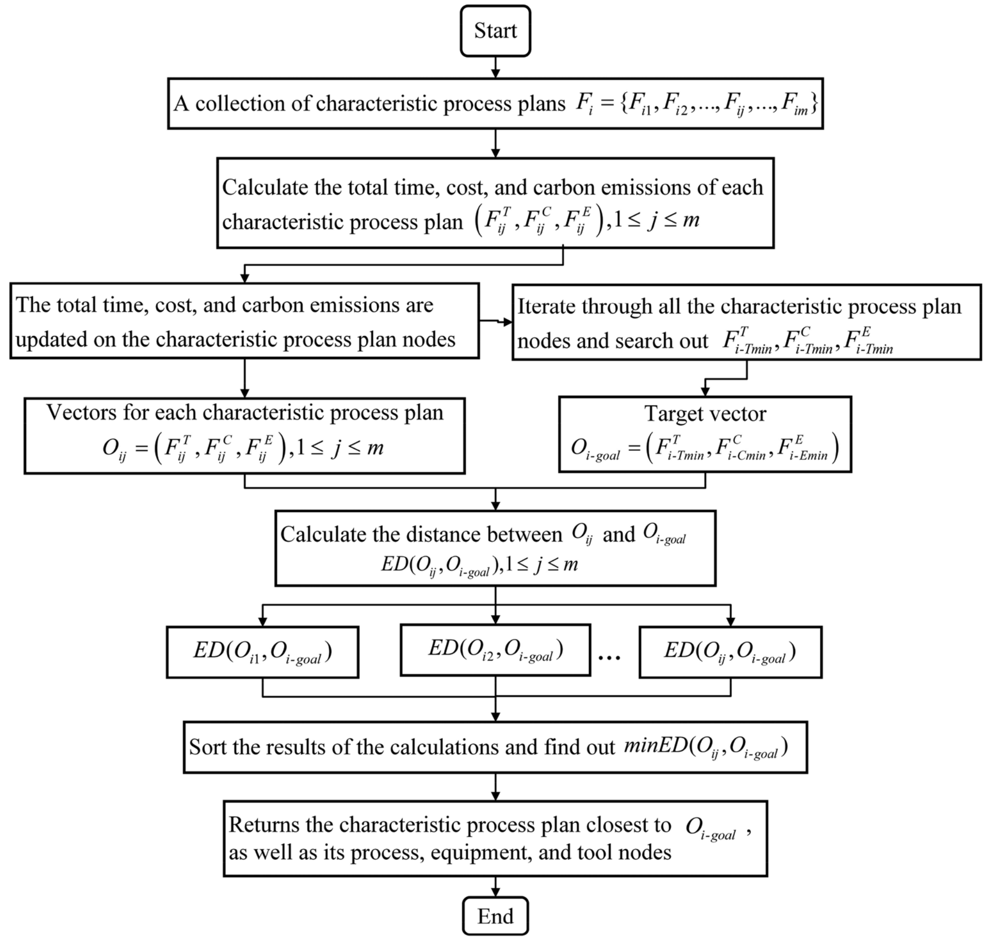

Figure 9. Best characteristic process plan search.

To ensure the entire retrieval process runs efficiently, Neo4j query performance is maintained by creating indexes and optimising query statements. Firstly, to reduce the frequency of full-graph traversals, three types of indexes are utilised to optimise Neo4j database performance: node label indexes (such as for parts and failure feature nodes), property indexes (such as for name and materials attributes), and composite indexes (for scenarios requiring simultaneous queries on name and materials). Secondly, writing efficient Cypher query statements is crucial for performance enhancement. When crafting query statements, wildcards are avoided, and node and relationship types are explicitly specified, effectively preventing full-graph searches and computations, thereby improving query efficiency.

5. Case study

Taking the failed worm gear as the research object, discusses the remanufacturing process planning method based on a knowledge graph, and verifies the effectiveness of the

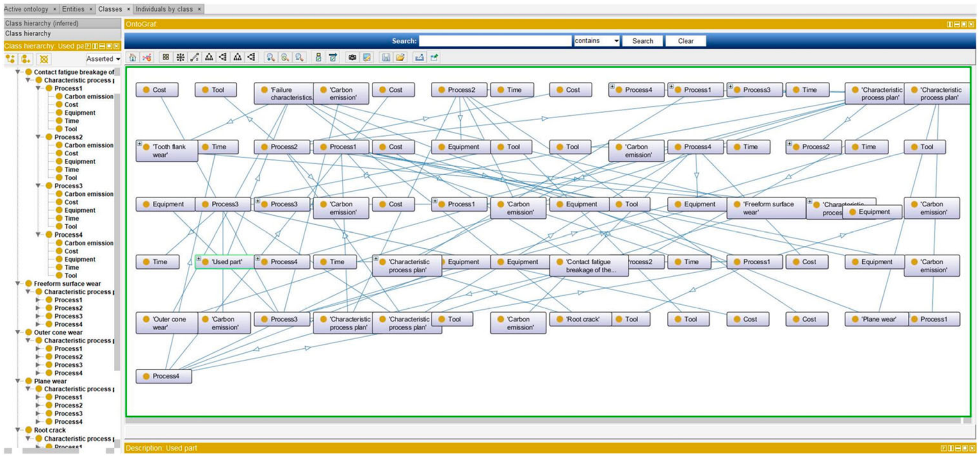

Figure 10. Ontology model of the remanufacturing process in Protégé.

proposed method. The worm gear drive is a mechanical device commonly used for power transmission and movement between two staggered axes. It has the advantages of self-locking, large transmission ratio, compact structure, low noise, and many other advantages. However, in the process of use, the worm gear often appears deformation, wear, cracks, and other failure problems. The production of new worm gears has a high cost, large carbon emission, and long processing time, and remanufacturing is more economical than direct production of new parts.

5.1. Construction of RPKG

To construct RPKG, partial historical data of different types of waste worm gears in a remanufacturing enterprise are collected. It includes the basic attribute characteristics, failure characteristics, process plan, and the time, cost, and carbon emission data generated in the process of waste worm gears. Firstly, according to the ontology-based remanufacturing process knowledge modelling method proposed in this paper, an ontology model including part characteristics, characteristic process plans, and key data indicators is established by using Protégé, as shown in Figure 10.

Then, the remanufacturing process data are divided into a training set and a test set at an 8:2 ratio. The training set is annotated according to the BIOES tagging scheme, and the annotated data are used to train the BERT-BiLSTM-CRF model. The model then automatically completes the entity extraction task for the remaining test set, resulting in the acquisition of 3,756 graph entities. The relationships between entities are defined by the relationships in the remanufacturing process ontology model. Finally, the RPKG, as shown in Figure 11, is constructed in Neo4j. In the BERT-BiLSTM-CRF model, the learning rate is set to 0.0001, the batch size is 32, the number of epochs is 20, the maximum length is 128, the hidden size is 192, and the drop rate is 0.1.

To validate the superiority of the proposed model, the performance of the entity extraction model is quantitatively evaluated using precision (P), recall rate (R), and F_1 value(Jiang et al. 2024). Comparative experiments based on the BiLSTM model and the BiLSTM-CRF model are designed. The results of entity extraction are shown in Table 6.

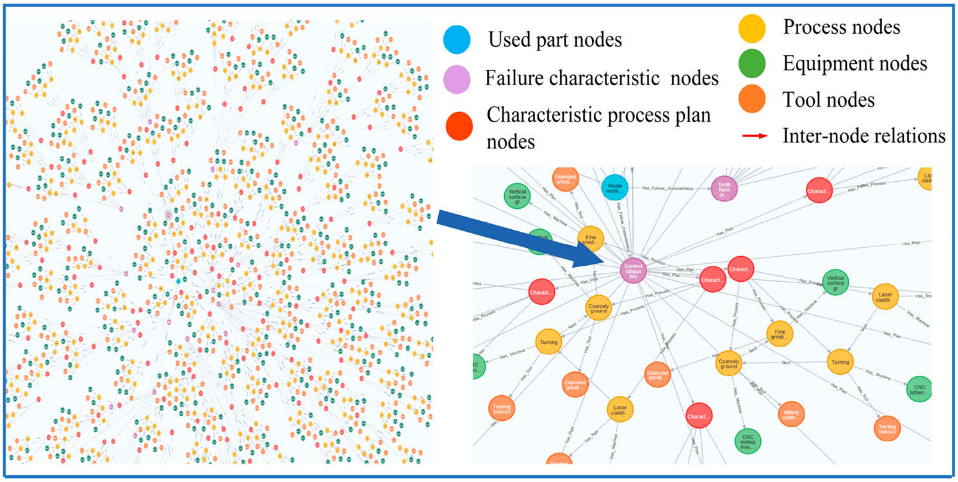

Figure 11. RPKG in the Neo4j.

Table 6. Comparison of entity extraction model results.

Model	Precision (*P*)	Recall rate (R)	F_1 value
BiLSTM	0.849	0.887	0.847
BiLSTM-CRF	0.858	0.872	0.856
BERT-BiLSTM-CRF	0.916	0.917	0.906

Table 7. Basic attribute characteristics of used worm gear A.

Basic attribute characteristics	Value	Basic attribute characteristics	Value
Module m/mm	4	Diameter factor q/mm	11.2
Indexing circle diameter d_1/mm	44.8	Correction factor x_2	+0.5
Pressure angle α/°	20	Material	C45
Number of worm gear heads z_1	2		

By comparing the precision, recall rate, and F_1 value of different models, the performance differences in the used part remanufacturing process entity extraction task can be observed. Specifically, although the BiLSTM model performs well, it is slightly inferior in terms of precision and F_1 value compared to the other two models. The BiLSTM-CRF model, with the introduction of the CRF layer, has its precision and F_1 value improved, but still remains lower than those of the BERT-BiLSTM-CRF model. The BERT-BiLSTM-CRF model, which combines a pre-trained BERT model, BiLSTM, and CRF, is characterised by the highest precision, recall, and F_1 value, demonstrating its superiority in the entity extraction task.

5.2. Characteristic matching of used parts

The basic attribute characteristics and failure characteristics of used worm gear A to be processed are shown in Table 7 and Table 8. According to the characteristics of used worm gear A, parts nodes and failure characteristic nodes that meet the similarity requirements are searched in RPKG.

Table 8. Failure characteristics of used worm gear A.

No	Failure characteristics	Volume of damage (mm^3)
S_{A1}	Contact fatigue breakage of the tooth surface	$\varphi = 0.87$
S_{A2}	Tooth flank wear	$\varphi = 0.85$
S_{A3}	Root crack	$\rho = 0.33$
S_{A4}	Freeform surface wear	$\varphi = 0.73$
S_{A5}	Plane wear	$\varphi = 0.67$
S_{A6}	Outer cone wear	$\varphi = 0.69$

Table 9. The basic attribute characteristics of each used worm gear.

Basic attribute characteristics	W_1	W_2	W_3	W_4	W_5
Module m/mm	4	4	5	2	5
Indexing circle diameter d_1 m/mm	40	71	50	22.4	90
Pressure angle α/°	20	20	20	20	20
Number of worm gear heads z_1	2	2	4	2	1
Diameter factor q/mm	10	17.75	10	11.2	17.75
Correction factor x_2	+0.75	+0.125	+1.0	+0.125	0
Material	C45	C45	C45	C45	C45

Table 10. Local similarity and global similarity results.

Similar used parts	W_1	W_2	W_3	W_4	W_5
Similarity of m	1.0	1.0	0.667	0.333	0.667
Similarity of d_1	0.929	0.929	0.923	0.669	0.331
Similarity of α	1.0	1.0	1.0	1.0	1.0
Similarity of z_1	1.0	1.0	0.333	1.0	0.667
Similarity of q	0.845	0.155	0.845	1.0	0.155
Similarity of x_1	0.75	0.625	0.5	0.625	0.5
Similarity of material	1.0	1.0	1.0	1.0	1.0
Global similarity	0.932	0.770	0.752	0.827	0.617

According to the basic attribute characteristics of the used parts to be processed, the used parts nodes that meet the similarity requirements are retrieved from RPKG in combination with algorithm 1. Table 9 shows the basic attribute characteristics of 5 waste worm gear nodes W_1 to W_5 found from RPKG. The local similarity and global similarity results for the basic attribute characteristics of W_1 to W_5 and the waste worm gear A to be processed are presented in Table 10. To ensure the feasibility of the remanufacturing process scheme, ϵ_1 is set to 0.9, so that only W_1 satisfies the similarity conditions.

After finding the waste parts node W_1 which is most similar to waste worm gear a, the failure characteristic node S_i directly connected to node W_1 is further searched. The similarity of each failure feature S_{Ai} between node S_i and worm gear A is calculated, and the failure feature node that satisfies the similarity condition ϵ_2 and is most similar to S_{Ai} is found. The most similar failure characteristic nodes $S_1 \sim S_6$ and their similarity are shown in Table 11. Figure 12 shows the search and matching process of used parts nodes and failure feature nodes.

5.3. Selection of optimal characteristic process plan and result analysis

After the feature node matching is completed, the retrieved failure characteristic node is taken as the starting node to start traversing, and the characteristic process plans of $S_1 \sim S_6$

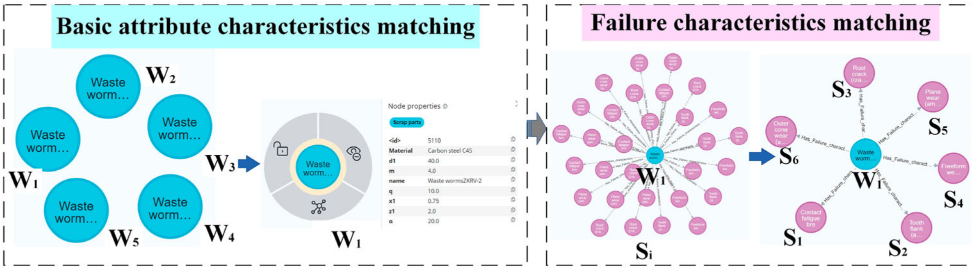

Figure 12. Feature nodes matching.

Table 11. The most similar failure characteristic nodes and their similarity.

Failure characteristic nodes	Failure characteristics	Local similarity	Volume of damage (mm^3)	Local similarity	Global similarity
S_1	Contact fatigue breakage of the tooth surface	1.0	$\varphi = 0.85$	0.980	0.990
S_2	Tooth flank wear	1.0	$\varphi = 0.88$	0.970	0.985
S_3	Root crack	1.0	$\rho = 0.34$	0.990	0.995
S_4	Freeform surface wear	1.0	$\varphi = 0.71$	0.980	0.990
S_5	Plane wear	1.0	$\varphi = 0.67$	1.0	1.0
S_6	Outer cone wear	1.0	$\varphi = 0.69$	1.0	1.0

are obtained. For each failure characteristic, there are several feasible characteristic process plans, as shown in Figure 13 (a), the details are shown in Table 12.

The NNS in Section 4.2 is used to retrieve the optimal process plan considering the total time, cost, and carbon emission for each failure characteristic of worm gear A, as shown in Figure 13 (b), detailed information is shown in Table 13.

To ensure a smooth and efficient remanufacturing process, the remanufacturing process plans for each failure characteristic are ranked according to the following constraint relations.

(1) Sequence constraint: The steps in each process plan must be executed in a specific order. For example, first turning and then grinding, and first roughing and then finishing. Additionally, surface treatment step such as brush plating is usually carried out before turning or rough grinding.
(2) Number of tool changes: Try to perform the steps using the same milling cutter in succession to minimise the number of tool changes. For example, after the steps using milling cutter a_5 are completed, replace it with milling cutter a_6 for fine grinding.

According to the above constraints, the process schemes of failure characteristics were sorted and combined, the repair sequence of each failure characteristic is $S_{A5} \rightarrow S_{A1} \rightarrow S_{A3} \rightarrow S_{A6} \rightarrow S_{A4} \rightarrow S_{A2}$. The process plan of worm gear A is obtained by combining the repeated processes. The total processing time of the scheme is 30.76 min, the cost is 24.99 yuan, and the carbon emission is 11.96kgCO2e.

To verify the effectiveness and superiority of the proposed method, the proposed method is compared with the remanufacturing process optimisation based on CBR and

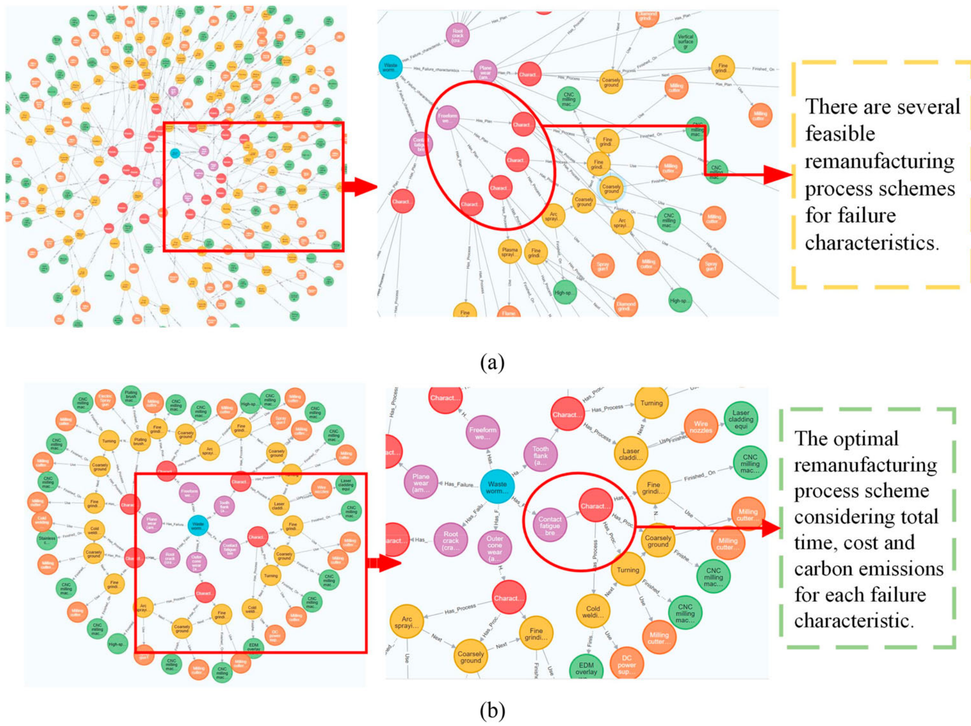

Figure 13. (a) Characteristic process plans for each failure characteristic; (b) The optimal characteristic process plan for each failure characteristic.

Genetic Algorithm (GA). The process plans based on the three methods are shown in Table 14.

The total time, cost, and carbon emissions of the three remanufacturing process options are shown in Table 15. The process scheme based on NNS-KG has the least processing time, cost, and carbon emissions. Additionally, this method reduces the need for complex models and algorithms, and lowers the demand for specialised skills and experience. For used parts with different failure characteristics, only the to-be-processed used parts and their attribute characteristics in the code need to be adjusted. The CBR-based approach solves new problems by matching historical cases, allowing similar solutions to be found quickly. However, suitable matching cases may not be found when unprecedented combinations of failure characteristics occur. The GA-based method can also achieve satisfactory results, but the remanufacturing process plan involves processing methods, equipment, tools, and processing sequences, and must take into account three constraints: time, cost, and carbon emissions. The encoding difficulty is relatively high, parameter optimisation is challenging, and the labour cost is considerable.

To validate the search efficiency and query performance of Neo4j, the entire search process is divided into four stages: Used parts node matching, Failure characteristic node matching, Generation of feasible solution set, and Search for optimal scheme. The search times for these four stages are recorded in Figure 14(a), and it can be observed that the search times are all at the millisecond level, indicating extremely high efficiency. Furthermore, to further verify the query performance of Neo4j, searches are conducted on various

Table 12. Detailed information for each characteristic process plan.

Failure feature node	Failure characteristics	Plan	Characteristic process plan
S_1	Contact fatigue breakage of the tooth surface ($\varphi = 0.85$)	F_{11}	Plating brush (Electroplating brush machine2) → Turning (CNC milling machine f_3, Milling cutter a_5) → Rough grinding (CNC milling machine f_3, Milling cutter a_5) → Fine grinding (CNC milling machine f_3, Milling cutter a_5)
		F_{12}	Cold welding (Stainless steel cold welding machine) → Turning (CNC lathes f_2, Turning tool a_3) → Rough grinding (CNC milling machine f_3, Milling cutter a_5) → Fine grinding (CNC milling machine f_3, Milling cutter a_6)
		…	
S_2	Tooth flank wear ($\varphi = 0.88$)	F_{21}	Flame spraying (Supersonic flame spray machine) → Rough grinding (CNC milling machine f_3, Milling cutter a_5) → Fine grinding (CNC milling machine f_3, Milling cutter a_5)
		F_{22}	Plating brush (Electroplating brush machine1) → Rough grinding (CNC milling machine f_3, Milling cutter a_5) → Fine grinding (CNC milling machine f_3, Milling cutter a_6)
		…	
S_3	Root crack ($\rho = 0.34$)	F_{31}	Cold welding (Electrospark surfacing cold welder) → Rough grinding (CNC milling machine f_3, Milling cutter a_5) → Fine grinding (CNC milling machine f_3, Milling cutter a_6)
		F_{32}	Cold welding (Stainless steel cold welding machine) → Rough grinding (CNC milling machine f_3, Milling cutter a_5) → Fine grinding (CNC milling machine f_3, Milling cutter a_5)
		…	
S_4	Freeform surface wear ($\varphi = 0.71$)	F_{41}	Arc spraying (High speed arc spraying machine) → Rough grinding (CNC milling machine f_3, Milling cutter a_5) → Fine grinding (CNC milling machine f_3, Milling cutter a_6)
		F_{42}	Arc spraying (High speed arc spraying machine) → Rough grinding (CNC milling machine f_3, Milling cutter a_5) → Rough grinding (Vertical surface grinding machine f_4, Diamond grinding wheels a_8)
		…	
S_5	Plane wear ($\varphi = 0.67$)	F_{51}	Arc spraying (High speed arc spraying machine) → Rough grinding (Vertical surface grinding machine f_4, Diamond grinding wheels a_8) → Fine grinding (CNC milling machine f_3, Milling cutter a_5) → Fine grinding (CNC milling machine f_3, Milling cutter a_6)
		F_{52}	Plating brush (Electroplating brush machine2) → Turning (CNC milling machine f_3, Milling cutter a_5) → Rough grinding (Vertical surface grinding machine f_4, Diamond grinding wheels a_8) → Fine grinding (CNC milling machine f_3, Milling cutter a_6)
		…	
S_6	Outer cone wear ($\varphi = 0.69$)	F_{61}	Plasma spraying (Plasma arc spraying machine) → Rough grinding (CNC milling machine f_3, Milling cutter a_5) → Fine grinding (CNC milling machine f_3, Milling cutter a_5)
		F_{62}	Arc spraying (High speed arc spraying machine) → Rough grinding (CNC milling machine f_3, Milling cutter a_5) → Fine grinding (CNC milling machine f_3, Milling cutter a_6)

remanufacturing process entities within the RPKG. The query times for these entities, as shown in Figure 14(b), are also at the millisecond level. The results demonstrate that the RPKG-based method exhibits extremely high search efficiency in practical applications, and the Neo4j graph database is noted for its excellent query performance.

6. Summary

To enhance the efficiency and quality of remanufacturing process planning for used parts with complex failure characteristics, a remanufacturing intelligent process planning method based on KG is proposed. Considering the variability in failure characteristics of

Table 13. Optimal process plan for each failure characteristic of worm gear A.

Failure characteristics of worm gear A	Failure characteristics	Characteristic process plan
S_{A1}	Contact fatigue breakage of the tooth surface ($\varphi = 0.87$)	Cold welding (Electrospark surfacing cold welder) → Turning (CNC milling machine f_3, Milling cutter a_5) → Rough grinding (CNC milling machine f_3, Milling cutter a_5) → Fine grinding (CNC milling machine f_3, Milling cutter a_6)
S_{A2}	Tooth flank wear ($\varphi = 0.85$)	Plating brush (Electroplating brush machine1) → Rough grinding (CNC milling machine f_3, Milling cutter a_5) → Fine grinding (CNC milling machine f_3, Milling cutter a_6)
S_{A3}	Root crack ($\rho = 0.33$)	Cold welding (Electrospark surfacing cold welder) → Rough grinding (CNC milling machine f_3, Milling cutter a_5) → Fine grinding (CNC milling machine f_3, Milling cutter a_6)
S_{A4}	Freeform surface wear ($\varphi = 0.73$)	Arc spraying (High speed arc spraying machine) → Rough grinding (CNC milling machine f_3, Milling cutter a_5) → Fine grinding (CNC milling machine f_3, Milling cutter a_6)
S_{A5}	Plane wear ($\varphi = 0.67$)	Plating brush (Electroplating brush machine1) → Turning (CNC milling machine f_3, Milling cutter a_5) → Rough grinding (CNC milling machine f_3, Milling cutter a_5) → Fine grinding (CNC milling machine f_3, Milling cutter a_6)
S_{A6}	Outer cone wear ($\varphi = 0.69$)	Arc spraying (High speed arc spraying machine) → Rough grinding (CNC milling machine f_3, Milling cutter a_5) → Fine grinding (CNC milling machine f_3, Milling cutter a_6)

Table 14. The process scheme obtained by three methods.

Method	Process scheme
Based on NNS-KG	Plating brush (Electroplating brush machine1) → Cold welding (Electrospark surfacing cold welder) → Arc spraying (High speed arc spraying machine) → Plating brush (Electroplating brush machine1) → Turning (CNC milling machine f_3, Milling cutter a_5) → Rough grinding (CNC milling machine f_3, Milling cutter a_5)→ Fine grinding (CNC milling machine f_3, Milling cutter a_6)
Based on CBR	Arc spraying (High speed arc spraying machine) → Cold welding (Stainless steel cold welding machine) → Arc spraying (High speed arc spraying machine) → Plating brush (Electroplating brush machine2) → Turning (CNC lathes f_2, Turning tool a_3) → Rough grinding (CNC milling machine f_3, Milling cutter a_5) → Fine grinding (CNC milling machine f_3, Milling cutter a_6)
Based on GA	Plating brush (Electroplating brush machine1) → Cold welding (Electrospark surfacing cold welder) → Plating brush (Electroplating brush machine2) → Turning (CNC lathes f_2, Turning tool a_3) → Rough grinding (CNC milling machine f_3, Milling cutter a_5)→ Fine grinding (CNC milling machine f_3, Milling cutter a_6)

Table 15. Comparison of the effect of the three schemes.

Indicator	Based on GA	Based on CBR	Based on NNS-KG	Percentage improved (GA/CBR)
Time (min)	31.63	33.19	30.76	2.75%/7.32%
Cost (yuan)	25.12	25.63	24.99	0.52%/2.60%
Carbon emission (kgCO2e)	12.83	14.96	11.96	6.78%/20.05%

used parts, the diversity of remanufacturing methods, and the complex non-linear relationships among remanufacturing process elements, a full-element RPKG is constructed, with failure characteristics serving as decision nodes. This graph integrates the initial features of used parts, failure characteristics, and characteristic process plans. It clearly represents the complex relationships between various remanufacturing process elements in historical cases, facilitating efficient querying, reasoning, and knowledge reuse. Then, an intelligent decision model based on graph multi-node path retrieval is constructed, aiming to minimise carbon emissions, time, and cost. Combining feature similarity calculations

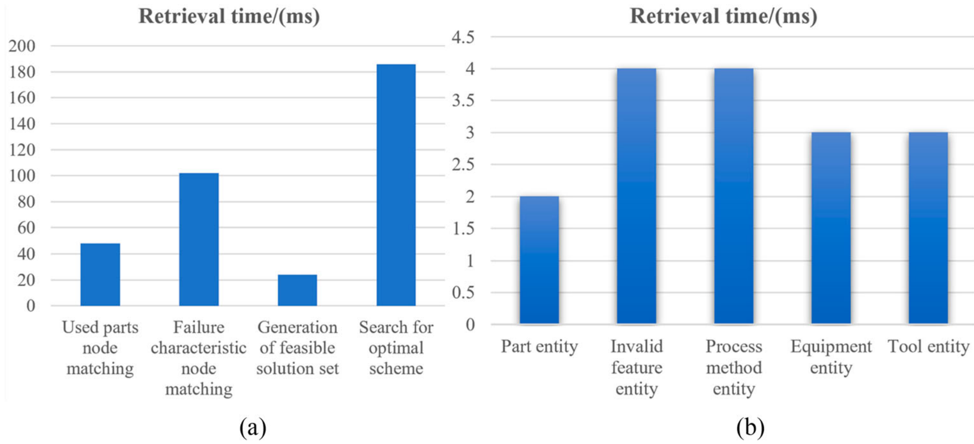

Figure 14. Search efficiency of each stage and query performance of Neo4j. (a) search time of the four stages; (b) search time of various entities.

and nearest neighbour search, the model efficiently and intelligently outputs the optimal remanufacturing process plans for each failure characteristic. Finally, according to the constraints between processes, the optimal remanufacturing process plans for each failure characteristic are integrated to obtain a complete remanufacturing process plan for used parts. The method effectively integrates the advantages of mathematical planning and knowledge reuse.

The BERT-BiLSTM-CRF model showed good performance in this study, but it is undeniable that it requires a lot of GPU memory and computing power to train when dealing with large-scale data sets containing millions or even billions of documents or sentences, and the training time can become longer. In addition, with the development of remanufacturing technology, one failure feature may correspond to more feasible solutions, and the search time for the best solution may increase. In the future, more efficient model architectures and training algorithms, such as the lightweight transformer variant, can be investigated to reduce the need for computing resources while maintaining performance. At the same time, more efficient graph traversal optimisation algorithms can be developed by combining specific heuristic functions to further improve the ability to process large-scale knowledge graphs.

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